Week 5

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2022-10-10

Where do I get slides/worksheets/solution?

Or go to fredazizi.github.io/Teaching

Quick review (1)

$P(A) \cdot P(C) = P(A \cap C)$

$P(A \mid B) \neq P(A \cap B)$

$P(A)$ $P(C)$

$P(B)$

$P(A)$ $P(B)$ $P(C \mid A)$ $P(D \mid B)$

- Probability Trees: Events in an experiment are represented by lines
- Joint distribution table $P(X \cap Y)$
- Marginal probabilites

$P(X = x, Y = y)$

		Y: C	Y: D	
X	A	$P(X=A, Y=C)$ $P(X=A \cap Y=C)$	$P(X=A, Y=D)$	$P(X=A)$
	B	$X=B, Y=C$	$P(X=B, Y=D)$	$P(X=B)$
		$P(Y=C) = P(X=A \cap Y=C) + P(X=B \cap Y=C)$		1

Quick review (2)

$x = 1, 2, 3, \ldots$

Fair Coin $\quad X$: number of H

X	0	1
P	$\frac{1}{2}$	$\frac{1}{2}$

- Discrete Probability distribution
  - $0 \leq P(x) \leq 1$ for all x
  - $\sum_{\text{all } x} P(x) = 1$
- Population Mean: $E(X) = \mu = \sum_{\text{all } x} xP(x)$
- Population Variance: $V(X) = \sigma^2 = \sum_{\text{all } x} (x - \mu)^2 P(x)$

$\bar{x} = \frac{\sum x}{n} =$

1	2	3
$\frac{1}{n}$	$\frac{1}{n}$	$\frac{1}{n}$

$SD(X) = \sqrt{\sigma^2}$

Quick review (3)

- Laws of Expected Value (note that c means a constant, a number)
  - $E(c) = c$
  - $E(X + c) = E(X) + c$
  - $E(cX) = cE(X)$
- Laws of Variance
  - $V(c) = 0$
  - $V(X + c) = V(X)$
  - $V(cX) = c^2 V(X)$